

Survey on LLM-as-a-Judge Reliability: A Review of Recent Advances

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Abstract

Large language models (LLMs) are increasingly used to evaluate open-ended model outputs, construct leaderboards, and select data or responses during training and inference. This shift creates a second-order reliability problem: an evaluator that is scalable and articulate may still be biased, inconsistent, poorly calibrated, contaminated, or vulnerable to manipulation. This survey synthesizes 24 recent papers through four complementary pillars: benchmarks and evaluation frameworks; bias, uncertainty, and fairness; scalability and adversarial robustness; and structured reasoning and judge training. The reviewed evidence spans pairwise preference, scalar scoring, system ranking, multilingual and multimodal assessment, social interaction, human-annotation replication, and test-time scaling. Across these settings, no single agreement score is sufficient to establish reliability. Judge quality depends on the task, evaluation format, model family, prompt and parser, data provenance, and relationship between the judge and evaluated model. Recent methods improve individual components through bias detection, uncertainty-aware training, selective escalation, checklists, crowd comparisons, and regression-aware chain-of-thought fine-tuning, yet they do not remove the need for human validation. We conclude with a reliability-oriented deployment framework that combines task-specific meta-evaluation, perturbation and contamination audits, repeated trials, calibrated abstention, and downstream utility tests.

1 Introduction

The rapid advancement of Large Language Models (LLMs) has made open-ended generation a central capability in modern AI systems. Evaluating such outputs is difficult because quality often depends on correctness, relevance, reasoning, style, safety, and task-specific constraints that cannot be captured by lexical-overlap metrics such as BLEU or ROUGE. Human evaluation remains the most direct way to measure these properties, but it is expensive, slow, sometimes inconsistent, and difficult to reproduce at leaderboard scale.

The “LLM-as-a-Judge” paradigm attempts to close this gap by asking a language model to apply a natural-language rubric and return a score, preference, critique, or ranking. Early results from MT-Bench and Chatbot Arena showed that strong judges can approximate human preferences in open-ended dialogue (Zheng et al., 2023). Since then, judges have moved beyond retrospective benchmarking: they now help construct preference data, rerank candidates, guide search, and refine outputs at test time. Consequently, a wrong verdict can affect not only a reported metric but also which models and data are selected for future development.

The central question is therefore not whether LLM judges are useful, but under what evidence they can be trusted. A high average agreement rate may coexist with position bias (Wang et al., 2024), run-to-run inconsistency (Halder & Hockenmaier, 2025), transferable adversarial attacks (Raina et al., 2024), or preference leakage from related synthetic-data generators (Li et al., 2026). Moreover, agreement on static pairwise benchmarks need not predict whether a judge can improve a generator during test-time scaling (Zhou et al., 2025).

This survey makes three contributions. First, it consolidates the 24-paper course corpus into a four-pillar reliability taxonomy that connects evaluation settings, failure modes, and mitigation strategies. Second, it

compares papers at the level of the evidence they provide—human agreement, stability, calibration, attack resistance, system ranking, and downstream utility—rather than treating all reported judge scores as interchangeable. Third, it derives a practical reliability protocol and identifies open problems in task-specific validation, uncertainty, contamination, adversarial security, and human escalation.

2 Background

An LLM judge can be described as a conditional evaluator J that receives an evaluation instruction or rubric r , a task input x , one or more candidate outputs y , and optionally a reference answer or auxiliary evidence e . It produces a judgment $z = J(r, x, y, e)$, where z may be a categorical label, pairwise preference, scalar score, ranking, or natural-language critique. Reliability concerns whether this output is accurate for the intended criterion, stable under irrelevant variation, calibrated to its probability of agreement, and useful in the downstream decision for which it is produced.

Evaluation formats. In *direct assessment*, the judge assigns a score or label to one response. In *pairwise comparison*, it selects between two responses or declares a tie. Repeated pairwise decisions can be aggregated into a system ranking, while generative judges may additionally produce critiques that guide response refinement or search. These formats are not equivalent: pairwise comparison often reduces scale-calibration demands, whereas absolute scoring is especially susceptible to transferred adversarial phrases (Raina et al., 2024); system ranking and test-time selection introduce aggregation and downstream-utility requirements (Gera et al., 2025; Zhou et al., 2025).

Reliability dimensions. Human agreement remains important, but the literature reveals at least five distinct dimensions: (1) validity against human or objective labels; (2) consistency across repetitions and semantically irrelevant perturbations; (3) calibration and uncertainty under distribution shift; (4) robustness to strategic manipulation and data contamination; and (5) utility for the downstream operation, such as system ranking, reranking, search, or refinement. Metrics for these dimensions are summarized in Appendix C.

Sources of error. Reliability failures can originate from the input order and style, the judge’s stochastic decoding, its training data and model family, the evaluation prompt and parser, or a mismatch between benchmark and deployment. Position, verbosity, authority, and self-preference biases are visible at the response level; preference leakage is a pipeline-level contamination effect; and universal adversarial phrases target the judge as an attack surface. This distinction matters because each failure requires a different audit and mitigation.

3 Methodology / Taxonomy

3.1 Scope and Corpus Construction

The corpus consists of 24 papers selected over four course-project phases, with six papers contributed in each phase. The selection is purposive rather than an exhaustive systematic review: it emphasizes influential benchmarks, empirical reliability audits, theoretical and adversarial analyses, and recent mitigation methods. We use the supplied paper PDFs as the primary source set and preserve each paper’s stated evaluation setting and motivation. The two sample surveys guide presentation—explicit scope, taxonomy, cross-paper comparison, limitations, and future directions—but are not included in the reviewed corpus.

Our synthesis is organized by three research questions: **RQ1:** In which tasks and formats do LLM judges align with human or objective evaluation? **RQ2:** Which sources of bias, uncertainty, contamination, and attack undermine this alignment? **RQ3:** Which training, prompting, calibration, and human-in-the-loop strategies improve reliability, and what limitations remain?

3.2 Reliability Taxonomy

Rather than listing papers chronologically, we use a four-pillar taxonomy based on the primary reliability question addressed by each work:

- **Category 1: Benchmarks & Frameworks for Judges.** These works define what judge quality means in settings ranging from dialogue and system ranking to multilingual, multimodal, broad NLP, social-interactive, and test-time-scaling evaluation.
- **Category 2: Bias, Uncertainty & Fairness Analysis.** These papers isolate response-level biases, run-to-run instability, uncertainty under distribution shift, and pipeline-level contamination, and evaluate mitigation without assuming that average agreement implies fairness.
- **Category 3: Scalability Limits & Adversarial Robustness.** These works study fine-tuned judge models, cost-aware design, theoretical limits, selective escalation, benchmark gaming, and transferable attacks against zero-shot assessment.
- **Category 4: Structured Reasoning & Judge Training.** These methods restructure evaluation through instance-specific checklists, crowd comparisons, or objectives that align chain-of-thought reasoning with numerical score prediction.

Table 1 summarizes the classification. Several papers span multiple axes; for example, JETTS is a benchmark but also tests downstream utility, while TRACT is both a training method and a reasoning method. Each paper is placed according to its central contribution, and cross-category implications are discussed later.

Category	Representative Papers	Primary Focus
Cat. 1: Benchmarks & Frameworks for Judges	Zheng et al. (2023); Tan et al. (2025); Gera et al. (2025); Zhou et al. (2024); Chen et al. (2024a); Pombal et al. (2025); Bavaresco et al. (2025); Zhou et al. (2025)	Meta-evaluation across dialogue, NLP, social, multilingual, multimodal, ranking, and test-time-scaling settings.
Cat. 2: Bias, Uncertainty & Fairness	Wang et al. (2024); Haldar & Hockenmaier (2025); Chen et al. (2024b); Ye et al. (2025); Yang et al. (2025); Xie et al. (2025); Li et al. (2026)	Measuring bias, self-inconsistency, model uncertainty, distribution shift, and preference leakage.
Cat. 3: Scalability & Adversarial Robustness	Zhu et al. (2025); Zheng et al. (2025); Raina et al. (2024); Dorner et al. (2025); Salinas et al. (2025); Jung et al. (2024)	Scaling judge deployment while exposing attacks, tuning cost/accuracy trade-offs, and deriving risk guarantees.
Cat. 4: Structured Reasoning & Judge Training	Wei et al. (2025); Zhang et al. (2025); Chiang et al. (2025)	Structuring evidence and aligning reasoning or training objectives with the final judgment.

Table 1: Taxonomic classification of the reviewed literature into four primary research branches.

This taxonomy answers the three research questions at different levels: Category 1 establishes validity across settings; Category 2 diagnoses non-adversarial failure; Category 3 tests operational limits and hostile conditions; and Category 4 improves the judgment process itself.

3.3 Evolutionary Trajectory of Reliability Research

The four categories describe the current structure of the literature, but the field has also undergone a recognizable conceptual progression. Figure 1 summarizes this trajectory. The stages overlap in time and should not be interpreted as a strictly linear sequence; rather, they identify changes in the dominant research question. Early work asked whether a strong LLM could approximate human preference at useful scale. Subsequent studies progressively shifted attention toward the conditions under which that agreement fails, the breadth of settings in which it can be reproduced, and the controls required when judge outputs influence model development.

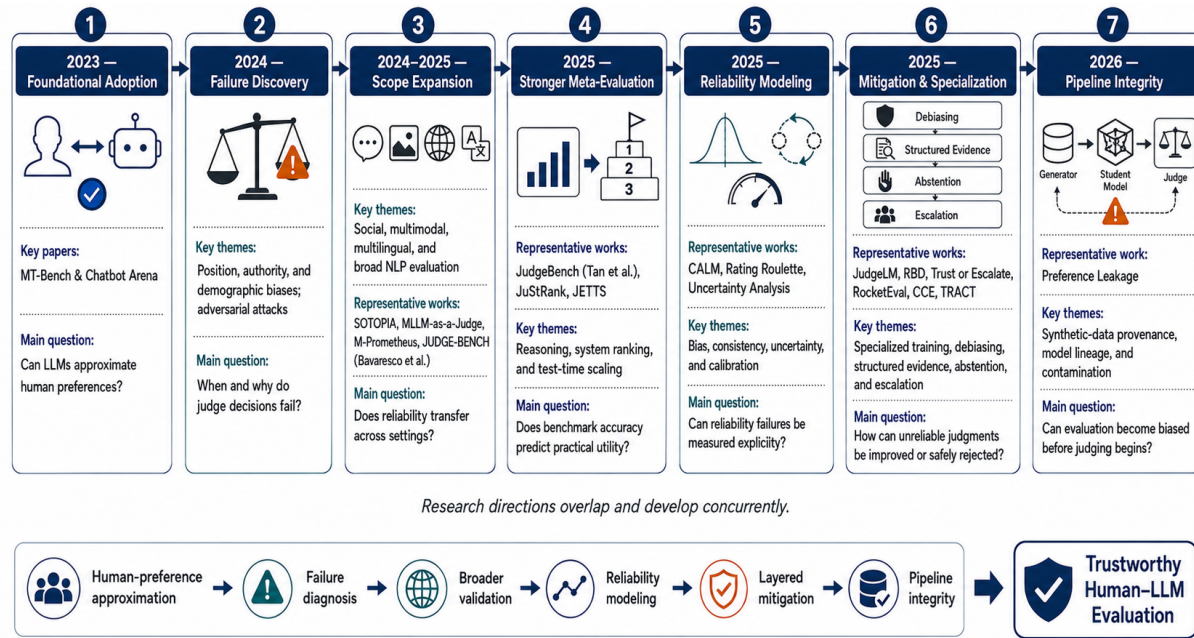


Figure 1: Evolutionary trajectory of LLM-as-a-Judge reliability research. The field progressed from approximating human preferences to diagnosing failures, validating judges across broader settings, modeling reliability explicitly, developing layered mitigation, and auditing pipeline-level contamination. The stages indicate dominant research emphases rather than a strictly chronological sequence.

Three broad trends emerge from this trajectory. First, the target of evaluation has expanded from isolated response preferences to model rankings and judge-guided inference, so reliability must increasingly be measured by downstream consequences rather than agreement alone. Second, the unit of analysis has widened from a single judge call to the complete pipeline: prompts, parsers, repeated sampling, training data, model lineage, and escalation rules can all influence the final decision. Third, mitigation has evolved from prompt-level corrections toward layered systems that combine specialized training, structured evidence, uncertainty estimation, adversarial testing, and selective human review. The field has therefore moved from asking *whether LLMs can judge* to asking *when their judgments are valid, what decisions they can safely support, and what evidence is required before they are trusted*.

4 Review of Key Papers

This section examines the literature according to the taxonomy above. A consolidated paper-level comparison of all 24 works is provided in Appendix A (Table 2), keeping the main discussion focused on thematic synthesis.

4.1 Benchmarks and Frameworks for Evaluating Judges

As the reliance on LLMs for automated evaluation has grown, so too has the need for standardized frameworks to assess the reliability of these automated judges. The foundational shift toward systematic benchmarking was established by Zheng et al. (2023), who introduced the *MT-Bench* and *Chatbot Arena* frameworks. *MT-Bench* provided a critical advancement by focusing on multi-turn conversational abilities, while *Chatbot Arena* introduced a crowdsourced, blind pairwise comparison platform serving as a real-world proxy for human preference.

Subsequent research has sought to create more rigorous and diverse testing environments. Tan et al. (2025) proposed *JudgeBench*, arguing that existing benchmarks over-index on stylistic alignment rather than objec-

tive reasoning. By curating challenging response pairs where one answer is objectively correct and the other contains subtle logical flaws, they demonstrated that LLM judges struggle when deep reasoning is required. Expanding into system-level evaluation, Gera et al. (2025) introduced *JuStRank*, revealing that a judge’s instance-level accuracy does not strictly correlate with its ability to reliably rank an entire ecosystem of models against a human-derived gold standard.

Beyond standard text evaluation, recent benchmarks have pushed into complex, interactive, and multimodal frontiers. Zhou et al. (2024) presented *SOTOPIA*, an open-ended environment designed to evaluate social intelligence through simulated, goal-driven interactions. Their framework revealed that while GPT-4 serves as a viable proxy for human judgment in certain dimensions, LLM agents still significantly lag behind humans in strategic communication and social goal completion. Simultaneously, Chen et al. (2024a) extended the evaluation paradigm to the vision-language domain with *MLLM-as-a-Judge*. They discovered that while multimodal judges align reasonably well with human preferences in pairwise comparisons, their reliability degrades significantly in scoring and batch ranking tasks due to vision-specific hallucinations and biases.

Pombal et al. (2025) addressed a major language-coverage gap with *M-Prometheus*, a suite of open multilingual judges trained for direct assessment and pairwise comparison across non-English outputs. The work shows that English-only reliability does not transfer automatically and that synthetic multilingual feedback can be more effective than simply translating English judge data. Language coverage, feedback language, and cross-lingual transfer are therefore reliability variables rather than implementation details.

Phase 4 broadens the meaning of a judge benchmark. Bavaresco et al. (2025) evaluate 11 open and proprietary LLMs on JUDGE-BENCH, an extensible collection of 20 NLP datasets and more than 70,000 instances with individual human annotations. Their cross-task results challenge universal claims: judges perform well for some properties, such as instruction following, but vary substantially with the task, data source, judgment scale, and annotator expertise, while ordinary chain-of-thought prompting does not reliably close the gap. The proper conclusion is task-specific validation, not wholesale replacement of human judges.

Zhou et al. (2025) go one step further by asking whether judge scores predict downstream usefulness. JETTS evaluates ten judges across response reranking, step-level beam search, and critique-based refinement in math, code, and instruction following. Judges can compete with outcome reward models for reranking, but trail process reward models during beam search, and their natural-language critiques are generally ineffective for refinement. The benchmark exposes a construct-validity gap: strong pairwise accuracy on RewardBench does not guarantee that a judge can select or improve responses sampled from the same generator. Together, JUDGE-BENCH and JETTS shift meta-evaluation from a single agreement number toward coverage across tasks and consequences.

4.2 Bias Quantification and Fairness Analysis

While benchmarks validate the general utility of automated judges, a critical line of research focuses on isolating their systemic flaws. Wang et al. (2024) argued that LLMs are not fair evaluators, exposing severe position bias where simply altering the order of candidate responses in the prompt can flip the judge’s verdict. To mitigate this, they proposed Multiple Evidence Calibration (MEC) and Balanced Position Calibration (BPC), forcing models to generate evidence prior to assigning scores.

Even when prompt positions are controlled, LLM judgments suffer from internal instability. Halder & Hockenmaier (2025) explored this through “Rating Roulette,” highlighting a lack of intra-rater reliability. Their experiments demonstrated that LLM judges can assign different scores to identical inputs across multiple identical runs, making single-call judgments volatile.

As these judges are deployed in high-stakes scenarios, uncovering deeper semantic prejudices has become critical. Chen et al. (2024b) developed a reference-free evaluation framework to quantify misinformation oversight, gender, authority, and beauty biases. Their study revealed that while human evaluators are susceptible to beauty and misinformation biases, cutting-edge LLMs suffer from all four to varying degrees and can be deceived by appending fake references to lower-quality answers. Further expanding this scope, Ye et al. (2025) introduced CALM, which systematically quantifies 12 distinct biases, including compassion-fade, band-

wagon, distraction, fallacy oversight, sentiment, diversity, chain-of-thought sensitivity, self-enhancement, and refinement-aware bias.

The next group moves from cataloguing bias toward modeling uncertainty and mitigation. Yang et al. (2025) proposed a Reasoning-based Bias Detector (RBD), a plug-in that inspects a judgment for verbosity, position, bandwagon, or sentiment bias before guiding revision. Because it does not modify the evaluator, the method is applicable to closed-source judges; results across eight evaluator configurations show gains in both accuracy and consistency.

Uncertainty is related to, but distinct from, observed self-inconsistency. Xie et al. (2025) analyze token-probability confidence for nine evaluators under pairwise comparison and single-answer grading. Uncertainty depends on model family and size, responds to prompt and post-training choices, and changes under distribution shift. Their uncertainty-aware evaluator, ConfiLM, uses confidence as auxiliary fine-tuning information and improves evaluation on a human-annotated out-of-distribution Olympics dataset. This result connects descriptive uncertainty to a deployment action: low confidence can inform training, failure detection, or escalation instead of being discarded.

Li et al. (2026) reveal a different, pipeline-level bias called *preference leakage*. When the model that generates synthetic training data is the same as, inherited by, or in the same family as the eventual judge, the judge systematically favors the resulting student model. The effect grows with generator–judge relatedness and the proportion of synthetic data, and is especially difficult to detect on subjective questions. Unlike ordinary self-preference, this contamination can persist even when model identities are hidden at evaluation time because the preference has already entered the student’s training distribution.

Together, this branch shows that substituting human evaluation with LLMs does not remove subjectivity or uncertainty. It introduces response-level, stochastic, model-level, and pipeline-level dependencies that must be measured separately.

4.3 Scalability Limits and Benchmark Robustness

As the adoption of LLM-as-a-Judge frameworks accelerates, critical operational challenges have emerged regarding scalability and robustness. Relying exclusively on proprietary models like GPT-4 is often cost-prohibitive. To address this bottleneck, Zhu et al. (2025) introduced *JudgeLM*, demonstrating that open-source LLMs can be fine-tuned on high-quality judgment datasets to serve as efficient and scalable judges, achieving over 90% agreement with teacher models while actively mitigating biases during training.

However, scaling automated evaluation exposes severe vulnerabilities. Zheng et al. (2025) demonstrate that adversarial “null models”—which ignore user instructions and output strategically structured constants—can exploit judge templates and parsers to obtain high win rates. This attack targets the complete evaluation pipeline rather than only the semantic competence of the judge.

Raina et al. (2024) attack the judge more directly. They optimize short universal phrases against a surrogate assessor, append those phrases to otherwise unchanged candidate text, and transfer them to unseen judge LLMs. The phrases can dramatically inflate scores across inputs and models, with absolute scoring more susceptible than comparative assessment. The result implies that robustness cannot be inferred from clean-set human correlation: deployed judges require adversarial evaluation, input sanitization, and format-specific defenses.

Cost and configuration choices further complicate judge deployment. Salinas et al. (2025) showed that judge quality is not determined by model scale alone: prompt strategy, output format, temperature, model choice, and parsing all interact. Their multi-objective, multi-fidelity tuning procedure searches thousands of judge configurations while trading off accuracy and evaluation cost, avoiding the prohibitive expense of full leaderboard-style evaluation for every configuration. This paper reframes judge design as an optimization problem rather than a fixed prompting recipe.

Even if security vulnerabilities are patched, theoretical ceilings exist on what automated judges can achieve. Dorner et al. (2025) provided a sobering statistical analysis on the limits of scalable evaluation at the frontier. Their theoretical proofs demonstrate that if a judge model is not strictly more accurate than the model it is

evaluating, optimal debiasing methods cannot decrease the required amount of ground-truth human labels by more than half. This proves that while LLM judges are highly effective for evaluating older models, they face hard mathematical limitations when deployed to benchmark frontier models that surpass their own capabilities.

Jung et al. (2024) provide a complementary route: instead of forcing every instance through a single judge, their selective evaluation framework calibrates when to trust, abstain, or escalate. Using simulated annotators and calibrated thresholds, the method can guarantee a target level of human agreement on evaluated examples, while cascaded evaluation starts with cheaper models and escalates only uncertain cases. This directly addresses a practical reliability question: not merely which judge is best on average, but when a given judge should be allowed to decide.

4.4 Structured Reasoning and Judge Training

The fourth pillar focuses on how evidence is generated and how it is connected to the final score. These methods are more than generic “think step by step” prompts. Wei et al. (2025) introduced *RocketEval*, which reframes judging as multi-faceted checklist grading. For each instance, the system creates a checklist, asks a lightweight judge to grade each item, and reweights checklist signals against supervised annotations. The method suggests that lightweight judges may be better at answering focused evaluative questions than at independently constructing a complete analysis.

Zhang et al. (2025) proposed *Crowd Comparative Evaluation*, which enriches pairwise judging by introducing additional crowd responses as comparison anchors. Instead of asking a judge to compare only response A and response B, the method first compares each candidate against several alternative responses, selects informative crowd judgments, removes explicit outcomes when needed, and then uses this evidence as context for the final judgment. This improves the comprehensiveness of chain-of-thought reasoning and yields consistent accuracy gains across multiple judge benchmarks. Conceptually, the method resembles human comparative review: exposure to a broader response pool makes subtle strengths and weaknesses easier to notice.

Chiang et al. (2025) address a complementary training problem for scalar judges. Standard cross-entropy treats predicting a score of 2 instead of 1 as no better than predicting 5 instead of 1, even though the numerical errors differ. TRACT combines chain-of-thought supervision with a regression-aware score loss and then retrains on self-generated reasoning to reduce the mismatch between externally supplied training rationales and inference-time rationales. Across multiple pointwise judge datasets and two base models, this objective improves Pearson and rank correlations over comparable baselines and transfers to pairwise RewardBench evaluation.

Together, RocketEval, Crowd Comparative Evaluation, and TRACT show that reliability depends on the interface between reasoning and decision. Decomposition can make evidence more complete, comparison anchors can reveal subtle differences, and regression-aware learning can align a numerical score with its evaluation metric. None guarantees faithful reasoning, however; intermediate text must still be evaluated for correctness and causal relevance.

5 Comparison and Discussion

5.1 What the Evidence Establishes

Across the 24 papers, four conclusions are well supported. First, judge validity is conditional rather than universal. Strong agreement in open-ended dialogue (Zheng et al., 2023) coexists with failures on subtle reasoning (Tan et al., 2025), variation across 20 NLP tasks (Bavaresco et al., 2025), and weak utility in some test-time-scaling settings (Zhou et al., 2025). Claims about a “reliable judge” must therefore name the task, population, format, and decision being supported.

Second, reliability is multi-dimensional. Average accuracy does not reveal position sensitivity (Wang et al., 2024), repeated-trial instability (Halder & Hockenmaier, 2025), confidence under distribution shift (Xie et al., 2025), or adversarial vulnerability (Raina et al., 2024; Zheng et al., 2025). Likewise, instance-level agreement

does not necessarily imply correct system ranking (Gera et al., 2025), and preference-benchmark accuracy does not ensure effective reranking or search (Zhou et al., 2025). Reporting several complementary metrics is not optional when the judge affects consequential choices.

Third, the evaluation format changes the attack surface. Pairwise comparison avoids some scalar-calibration difficulties and is more robust than absolute scoring to universal suffix attacks (Raina et al., 2024), but it remains vulnerable to response order, tie policy, and aggregation. Direct assessment supports fine-grained rubrics and critiques but introduces ordinal or regression errors, motivating checklist decomposition (Wei et al., 2025) and regression-aware learning (Chiang et al., 2025).

Fourth, judge independence matters. Self-preference has long been treated as a response-level bias, but preference leakage shows that dependence can enter earlier through synthetic-data generation (Li et al., 2026). A judge may favor a related student even when evaluator prompts are neutral. Reliable experiments should therefore document the model lineage and provenance of training, reference, and evaluation data.

5.2 Mitigation Trade-offs

The mitigation literature divides into four layers. Prompt and evidence methods, such as MEC/BPC, RocketEval, and crowd comparative reasoning, are easy to add but increase inference cost and can introduce new anchoring effects. External detectors such as RBD work with closed models, although their own reasoning must be validated. Fine-tuned judges such as JudgeLM, M-Prometheus, ConfiLM, and TRACT provide greater control but inherit their teachers, annotations, and training distributions. Selective and cascaded evaluation makes uncertainty operational by abstaining or escalating (Jung et al., 2024), but trades coverage and cost for stronger guarantees.

These approaches are complementary. A practical system can use a task-tuned judge, structured evidence, repeated order-balanced trials, calibrated confidence, and a human escalation rule. The important design shift is to treat reliability as an end-to-end property of data, prompts, models, parsing, aggregation, and decisions rather than as a property of the backbone alone.

5.3 A Reliability-Oriented Deployment Protocol

The synthesis suggests a five-stage protocol. **(1) Define the construct:** specify the quality dimensions, rubric, output format, and downstream decision. **(2) Validate locally:** compare against task-specific expert or crowd judgments and report human–human agreement. **(3) Stress-test:** repeat trials, swap candidate order, control length and style, test relevant bias perturbations, and run adversarial suffix and parser checks. **(4) Audit dependence:** record generator, student, reference, and judge lineages and test cross-family judges for preference leakage. **(5) Calibrate action:** report uncertainty, coverage, cost, and downstream utility; abstain or escalate when evidence is insufficient. This protocol is stricter than selecting the model with the highest benchmark accuracy, but it matches the range of failures observed in the corpus.

6 Open Problems and Future Directions

Task-conditional meta-evaluation. The breadth of JUDGE-BENCH and the operational design of JETTS should be combined: future benchmarks should measure both agreement on diverse tasks and the consequences of using a judge for selection, search, refinement, or training. Multilingual and multimodal work should additionally report results by language, culture, modality, and task rather than only macro averages (Pombal et al., 2025; Chen et al., 2024a).

Uncertainty with guarantees under shift. Token probabilities, verbalized confidence, and self-consistency are imperfect uncertainty signals. Future work should compare them under temporal and domain shift, calibrate them against human disagreement, and integrate them with selective-risk guarantees (Xie et al., 2025; Jung et al., 2024). A useful judge should know when its rubric or knowledge is insufficient, not merely when its next-token distribution is diffuse.

Contamination-aware evaluation. Preference leakage creates a need for model-lineage disclosure and cross-family evaluation. Open problems include black-box detection when training data are unavailable, separating legitimate stylistic similarity from leaked preference, and designing judge panels whose errors are genuinely diverse rather than inherited from a shared teacher (Li et al., 2026).

Adaptive adversarial security. Static perturbation sets may quickly become obsolete. Future benchmarks should include adaptive attackers, semantic-preservation checks, hidden prompt variants, parser fuzzing, and attacks against multi-judge or checklist systems (Raina et al., 2024; Zheng et al., 2025). Defenses must be evaluated against attack transfer, not only the attack used during development.

Faithful and metric-aware reasoning. Checklists, comparative context, and CoT-aware objectives improve outcomes, but generated rationales may still be post-hoc. Future work should test whether intermediate evidence causally supports the verdict, whether it is factually correct, and whether optimization for correlation preserves calibration and fairness (Wei et al., 2025; Zhang et al., 2025; Chiang et al., 2025).

Human–LLM evaluation design. Human evaluation should be allocated where it provides the most information: ambiguous items, distribution shifts, high disagreement, and cases near consequential thresholds. The field still needs principled methods for choosing the calibration set, estimating the required number of labels, and updating escalation policies as models and task distributions change (Dorner et al., 2025; Jung et al., 2024).

7 Conclusion

This survey reviewed 24 papers on LLM-as-a-Judge reliability through four pillars: benchmarks and frameworks; bias, uncertainty, and fairness; scalability and adversarial robustness; and structured reasoning and judge training. The literature supports a balanced conclusion. LLM judges are flexible and often economical evaluators, but agreement on a convenient benchmark is not evidence of general reliability. Their decisions depend on format, task, stochasticity, confidence calibration, model and data lineage, prompt and parser design, and adversarial conditions. Trustworthy use therefore requires a validated evaluation pipeline: define the construct, compare against task-specific humans, audit bias and contamination, stress-test attacks and repeated trials, measure downstream utility, and escalate uncertain cases. Under these controls, LLM judges can complement human evaluation; without them, scale risks amplifying measurement error.

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References

- Anna Bavaresco, Raffaella Bernardi, Leonardo Bertolazzi, Desmond Elliott, Raquel Fernández, Albert Gatt, Esam Ghaleb, Mario Giulianelli, Michael Hanna, Alexander Koller, André F. T. Martins, Philipp Mondorf, Vera Neplenbroek, Sandro Pezzelle, Barbara Plank, David Schlangen, Alessandro Suglia, Aditya K Surikuchi, Ece Takmaz, and Alberto Testoni. LLMs instead of human judges? a large scale empirical study across 20 NLP evaluation tasks. *arXiv preprint arXiv:2406.18403*, 2025.
- Dongping Chen, Ruoxi Chen, Shilin Zhang, Yinuo Liu, Huichi Zhou, Qihui Zhang, Yaochen Wang, Yao Wan, Pan Zhou, and Lichao Sun. MLLM-as-a-judge: Assessing multimodal LLM-as-a-judge with vision-language benchmark. In *International Conference on Machine Learning*, 2024a.
- Guiming Hardy Chen, Shunian Chen, Ziche Liu, Feng Jiang, and Benyou Wang. Humans or LLMs as the judge? A study on judgement biases. *arXiv preprint arXiv:2402.10669*, 2024b.
- Cheng-Han Chiang, Hung-yi Lee, and Michal Lukasik. TRACT: Regression-aware fine-tuning meets chain-of-thought reasoning for LLM-as-a-judge. *arXiv preprint arXiv:2503.04381*, 2025.
- Florian E. Dorner, Vivian Y. Nastl, and Moritz Hardt. Limits to scalable evaluation at the frontier: LLM as judge won’t beat twice the data. *arXiv preprint arXiv:2410.13341*, 2025.

- Ariel Gera, Odellia Boni, Yotam Perlitz, Roy Bar-Haim, Lilach Eden, and Asaf Yehudai. JuStRank: Benchmarking LLM judges for system ranking. In *Proceedings of the Association for Computational Linguistics*, 2025.
- Rajarshi Haldar and Julia Hockenmaier. Rating roulette: Self-inconsistency in LLM-as-a-judge frameworks. *arXiv preprint arXiv:2510.27106*, 2025.
- Jaehun Jung, Faeze Brahman, and Yejin Choi. Trust or escalate: LLM judges with provable guarantees for human agreement. *arXiv preprint arXiv:2407.18370*, 2024.
- Dawei Li, Renliang Sun, Yue Huang, Ming Zhong, Bohan Jiang, Jiawei Han, Xiangliang Zhang, Wei Wang, and Huan Liu. Preference leakage: A contamination problem in LLM-as-a-judge. In *International Conference on Learning Representations*, 2026.
- Jose Pombal, Dongkeun Yoon, Patrick Fernandes, Ian Wu, Seungone Kim, Ricardo Rei, Graham Neubig, and Andre F. T. Martins. M-Prometheus: A suite of open multilingual LLM judges. In *Conference on Language Modeling*, 2025.
- Vyas Raina, Adian Liusie, and Mark Gales. Is LLM-as-a-judge robust? investigating universal adversarial attacks on zero-shot LLM assessment. *arXiv preprint arXiv:2402.14016*, 2024.
- David Salinas, Omar Swelam, and Frank Hutter. Tuning LLM judge design decisions for 1/1000 of the cost. In *International Conference on Machine Learning*, 2025.
- Sijun Tan, Siyuan Zhuang, Kyle Montgomery, William Y. Tang, Alejandro Cuadron, Chenguang Wang, Raluca Ada Popa, and Ion Stoica. JudgeBench: A benchmark for evaluating LLM-based judges. In *International Conference on Learning Representations*, 2025.
- Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. In *Proceedings of the Association for Computational Linguistics*, 2024.
- Tianjun Wei, Wei Wen, Ruizhi Qiao, Xing Sun, and Jianghong Ma. RocketEval: Efficient automated LLM evaluation via grading checklist. In *International Conference on Learning Representations*, 2025.
- Qiuji Xie, Qingqiu Li, Zhuohao Yu, Yuejie Zhang, Yue Zhang, and Linyi Yang. An empirical analysis of uncertainty in large language model evaluations. In *International Conference on Learning Representations*, 2025.
- Haoyan Yang, Runxue Bao, Cao Xiao, Jun Ma, Parminder Bhatia, Shangqian Gao, and Taha Kass-Hout. Any large language model can be a reliable judge: Debiasing with a reasoning-based bias detector. In *Advances in Neural Information Processing Systems*, 2025.
- Jiayi Ye, Yanbo Wang, Yue Huang, Dongping Chen, Qihui Zhang, Nuno Moniz, Tian Gao, Werner Geyer, Chao Huang, Pin-Yu Chen, Nitesh V. Chawla, and Xiangliang Zhang. Justice or prejudice? quantifying biases in LLM-as-a-judge. In *International Conference on Learning Representations*, 2025.
- Qiyuan Zhang, Yufei Wang, Yuxin Jiang, Liangyou Li, Chuhan Wu, Yasheng Wang, Xin Jiang, Lifeng Shang, Ruiming Tang, Fuyuan Lyu, and Chen Ma. Crowd comparative reasoning: Unlocking comprehensive evaluations for LLM-as-a-judge. In *Proceedings of the Association for Computational Linguistics*, 2025.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Zi Lin, Zhuohan Li, Eric P. Xing, Joseph E. Gonzalez, Ion Stoica, and Hao Zhang. Judging LLM-as-a-Judge with MT-bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, 2023.
- Xiaosen Zheng, Tianyu Pang, Chao Du, Qian Liu, Jing Jiang, and Min Lin. Cheating automatic LLM benchmarks: Null models achieve high win rates. In *International Conference on Learning Representations*, 2025.

Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Zhengyang Qi, Haoifei Yu, Yonatan Bisk, Daniel Fried, Graham Neubig, Louis-Philippe Morency, and Maarten Sap. SOTOPIA: Interactive evaluation for social intelligence in language agents. In *International Conference on Learning Representations*, 2024.

Yilun Zhou, Austin Xu, Peifeng Wang, Caiming Xiong, and Shafiq Joty. Evaluating judges as evaluators: The JETTS benchmark of LLM-as-judges as test-time scaling evaluators. In *Proceedings of the International Conference on Machine Learning*, 2025.

Lianghui Zhu, Xinggang Wang, and Xinlong Wang. JudgeLM: Fine-tuned large language models are scalable judges. In *International Conference on Learning Representations*, 2025.

A Paper-Level Comparison

Table 2 summarizes the central contribution and principal reliability finding of every paper in the reviewed corpus.

Table 2: Comparative overview of the reviewed LLM-as-a-Judge reliability literature.

Paper (Venue, Year)	Cat.	Core Contribution	Key Finding
Zheng et al. (2023) (NeurIPS, 2023)	C1	Introduced MT-Bench and Chatbot Arena	GPT-4-based judges show strong agreement with human preference in open-ended, multi-turn dialogue.
Tan et al. (2025) (ICLR, 2025)	C1	JudgeBench: response pairs with one objectively correct, one subtly flawed answer	Judge reliability falls sharply on tasks requiring deep reasoning rather than surface style.
Gera et al. (2025) (ACL, 2025)	C1	JuStRank: system-level ranking benchmark	Instance-level judging accuracy does not reliably predict a judge’s ability to rank entire model ecosystems.
Zhou et al. (2024) (ICLR, 2024)	C1	SOTOPIA: interactive social-intelligence evaluation environment	GPT-4 is a partial proxy for human judgment, but agents still lag humans in strategic social goal completion.
Chen et al. (2024a) (ICML, 2024)	C1	MLLM-as-a-Judge: vision-language judge benchmark	Multimodal judges align well in pairwise comparison but degrade sharply in scoring and batch ranking.
Pombal et al. (2025) (COLM, 2025)	C1	Open multilingual judge suite from 3B to 14B parameters	Synthetic multilingual feedback data and backbone choice are crucial for multilingual DA and pairwise judging.
Bavaresco et al. (2025) (arXiv, 2025)	C1	JUDGE-BENCH collection of 20 NLP tasks and more than 70,000 human judgments	Reliability varies strongly by task, evaluated property, data source, and annotator expertise; task-specific human validation is necessary.
Zhou et al. (2025) (ICML, 2025)	C1	JETTS benchmark for reranking, beam search, and critique-based refinement	Static preference accuracy overstates operational utility: weak judges struggle on reasoning-intensive search and critiques rarely improve responses.
Wang et al. (2024) (ACL, 2024)	C2	Multiple Evidence Calibration (MEC) and Balanced Position Calibration (BPC)	Severe positional bias: reordering candidate responses alone can flip the judge’s verdict.
Halдар & Hockenmaier (2025) (arXiv, 2025)	C2	“Rating Roulette”: repeated-trial self-consistency analysis	LLM judges show low intra-rater reliability, assigning different scores to identical inputs across runs.
Chen et al. (2024b) (arXiv, 2024)	C2	Reference-free framework quantifying misinformation, gender, authority, and beauty biases	LLMs are susceptible to all four biases and can be fooled by appending fake references to weak answers.

Table 2 – continued

Paper (Venue, Year)	Cat.	Core Contribution	Key Finding
Ye et al. (2025) (ICLR, 2025)	C2	CALM: automated-perturbation framework covering 12 distinct biases	Confirms pervasive, multi-dimensional bias beyond position and verbosity alone.
Yang et al. (2025) (NeurIPS, 2025)	C2	Reasoning-based Bias Detector (RBD) plug-in	External bias detection and self-correction improve accuracy and consistency across evaluator families.
Xie et al. (2025) (ICLR, 2025)	C2	Uncertainty analysis of nine judges and the uncertainty-aware ConfiLM evaluator	Confidence varies by model family, scale, prompt, and distribution; using uncertainty during fine-tuning improves out-of-distribution evaluation.
Li et al. (2026) (ICLR, 2026)	C2	Defines and measures preference leakage between data generators and judges	Related judges systematically favor student models trained on their synthetic data, especially with greater relatedness and synthetic-data proportion.
Zhu et al. (2025) (ICLR, 2025)	C3	JudgeLM: fine-tuned open-source judge models	Achieves over 90% agreement with teacher judgments while mitigating bias via training-time augmentation.
Zheng et al. (2025) (ICLR, 2025)	C3	Adversarial “null model” attack on automatic benchmarks	Constant, instruction-ignoring outputs with an optimized prefix can game leaderboards via template exploitation.
Raina et al. (2024) (arXiv, 2024)	C3	Transferable universal adversarial phrases for zero-shot LLM assessment	Short suffixes learned on a surrogate can inflate unseen judges’ scores; absolute scoring is markedly more vulnerable than comparison.
Dorner et al. (2025) (arXiv, 2025)	C3	Statistical bound on scalable LLM-judge evaluation	A judge that is not strictly more accurate than the model it evaluates cannot cut required human labels by more than half.
Salinas et al. (2025) (ICML, 2025)	C3	Multi-objective, multi-fidelity judge hyperparameter tuning	Prompt, model, temperature, and parser choices can be optimized at roughly 1/1000 of exhaustive evaluation cost.
Jung et al. (2024) (arXiv, 2024)	C3	Selective and cascaded evaluation with risk guarantees	Calibrated abstention and escalation can guarantee target human agreement while using cheaper judges on many cases.
Wei et al. (2025) (ICLR, 2025)	C4	RocketEval: instance-specific grading checklist pipeline	Checklist creation, grading, and score reweighting let lightweight judges approach GPT-4o-level system ranking at much lower cost.

Table 2 – continued

Paper (Venue, Year)	Cat.	Core Contribution	Key Finding
Zhang et al. (2025) (ACL, 2025)	C4	Crowd Comparative Evaluation (CCE)	Comparing candidates against additional crowd responses yields richer CoT judgments and improves accuracy across judge benchmarks.
Chiang et al. (2025) (arXiv, 2025)	C4	TRACT combines regression-aware score learning with self-generated chain-of-thought	Matching the training objective to numeric error and reducing train–inference CoT mismatch improves correlation across pointwise judge datasets.

B Evaluation Formats

Table 3 summarizes the evaluation formats used across the 24-paper corpus. These formats differ in what the judge sees, what output it must produce, and what reliability risk is most important to measure.

Table 3: Major evaluation formats used in LLM-as-a-Judge research.

Format	Judge Input and Output	Representative Papers	Main Reliability Concern
Single-answer direct assessment	The judge receives one instruction-response pair, optionally with a rubric or reference, and assigns a scalar score or textual feedback.	Pombal et al. (2025); Wei et al. (2025); Xie et al. (2025); Chiang et al. (2025)	Scores can be poorly calibrated, unstable across runs, and sensitive to rubric wording.
Pairwise comparison	The judge receives one prompt and two candidate responses, then selects A, B, or tie.	Zheng et al. (2023); Wang et al. (2024); Xie et al. (2025); Raina et al. (2024)	Position bias, tie handling, parser fragility, and order sensitivity.
Multi-turn dialogue judging	The judge evaluates responses over conversation history rather than a single isolated turn.	Zheng et al. (2023); Zhou et al. (2024)	Long context can hide earlier errors; later-turn style may dominate instruction fidelity.
Objective correctness pair judging	One response is objectively correct and the other contains subtle factual, logical, mathematical, or coding errors.	Tan et al. (2025)	Human preference can reward plausibility or style over correctness; strong reasoning is required.
System-level ranking	Instance-level judgments are aggregated into a model ranking, leaderboard, or system ordering.	Gera et al. (2025); Salinas et al. (2025); Dorner et al. (2025)	Accurate local judgments may not induce reliable global rankings.

Table 3 – continued

Format	Judge Input and Output	Representative Papers	Main Reliability Concern
Human-annotation replication	The LLM reproduces categorical, ordinal, or scalar annotations originally collected from experts or non-experts across heterogeneous NLP tasks.	Bavaresco et al. (2025)	Performance varies with the property, task, data source, scale, and annotator expertise.
Crowdsourced battle / arena format	Humans or LLMs compare anonymous model responses in open-ended user-chatbot interactions.	Zheng et al. (2023); Dorner et al. (2025); Jung et al. (2024)	Real user preference is noisy, subjective, and expensive to collect.
Interactive social simulation	Agents interact in goal-driven social scenarios and are judged on social intelligence dimensions.	Zhou et al. (2024)	Outcome quality depends on dialogue dynamics, social goals, and subjective norms.
Multimodal judging	The judge evaluates image-text or vision-language outputs, using pairwise, scoring, or ranking formats.	Chen et al. (2024a)	Visual grounding errors and hallucinations interact with ordinary text-judge biases.
Multilingual direct assessment and pairwise comparison	The judge evaluates non-English outputs and may produce feedback or pairwise verdicts across many languages.	Pombal et al. (2025)	English-centric judge training may not transfer; translation-based training can distort evaluation behavior.
Bias perturbation format	A control response and a bias-injected variant are compared to test whether irrelevant changes alter the verdict.	Chen et al. (2024b); Ye et al. (2025); Yang et al. (2025); Li et al. (2026)	The perturbation must preserve semantic quality while isolating one bias trigger or lineage relationship.
Repeated-trial scoring	The same judge evaluates the same item multiple times under nominally identical settings.	Halder & Hockenmaier (2025)	Low intra-rater reliability makes one-shot scores unreliable.
Uncertainty-aware evaluation	The judge produces a verdict together with confidence derived from output probabilities, repeated samples, or a calibrated model.	Xie et al. (2025); Jung et al. (2024)	Raw confidence may be miscalibrated and sensitive to domain shift.

Table 3 – continued

Format	Judge Input and Output	Representative Papers	Main Reliability Concern
Checklist-based grading	The response is decomposed into instance-specific checklist items, each graded separately before aggregation.	Wei et al. (2025)	Checklist generation quality and item weighting determine final alignment.
Crowd comparative reasoning	Candidate responses are compared against additional crowd responses before the final A/B judgment.	Zhang et al. (2025)	Added context must expose useful details without introducing new anchoring bias.
Fine-tuned judge evaluation	A smaller or open model is trained to output judge decisions or scores using teacher or preference data.	Zhu et al. (2025); Pombal et al. (2025)	Training data can encode teacher bias and may fail under distribution shift.
Test-time-scaling evaluation	A judge reranks complete candidates, guides step-level search, or critiques an answer for refinement.	Zhou et al. (2025)	Static benchmark accuracy may not predict improvement over greedy generation or reward-model baselines.
Selective / cascaded judging	A judge outputs a verdict only when calibrated confidence is high; uncertain cases are escalated or abstained.	Jung et al. (2024)	Reliability depends on calibrated confidence, coverage, and escalation cost.
Adversarial benchmark-gaming format	A model submits strategically formatted outputs or universal phrases that exploit the judge prompt, learned preference, or parser.	Zheng et al. (2025); Raina et al. (2024)	Scores may reflect template or model exploitation rather than output quality.

C Alignment Metrics

Table 4 lists the major metrics used to quantify whether an LLM judge is aligned with human judgment, objective correctness, or a desired reliability constraint. A robust evaluation should normally report several of these metrics together, because high average agreement can coexist with bias, poor calibration, or low consistency.

Table 4: Alignment and reliability metrics used for LLM-as-a-Judge systems.

Metric	Definition / Use	Representative Papers	What It Misses
Human agreement rate	Fraction of examples where the LLM judge matches a human majority or expert label.	Zheng et al. (2023); Tan et al. (2025); Bavaresco et al. (2025); Jung et al. (2024)	Can hide subgroup bias, calibration error, and disagreement severity.
Human-human agreement baseline	Agreement among human annotators, used as a practical ceiling or comparison point for LLM judges.	Zheng et al. (2023); Jung et al. (2024)	Humans can share systematic biases and may be unreliable on expert tasks.
Pairwise accuracy	Accuracy of A/B/tie preferences against gold preference labels or objective labels.	Tan et al. (2025); Zhang et al. (2025); Yang et al. (2025)	Treats all errors equally and may ignore confidence.
Win rate / tie rate	Proportion of pairwise battles won, lost, or tied by a system; often used in arena-style comparisons.	Zheng et al. (2023); Zheng et al. (2025); Pombal et al. (2025)	Vulnerable to prompt mix, opponent pool, and parser manipulation.
Elo / Bradley-Terry style rating	Converts many pairwise battles into a global model score or ranking.	Zheng et al. (2023); Gera et al. (2025); Dorner et al. (2025)	Assumes pairwise outcomes aggregate cleanly into stable rankings.
Spearman rank correlation	Measures correlation between judge-induced system ranking and human or gold ranking.	Salinas et al. (2025); Wei et al. (2025); Dorner et al. (2025)	Insensitive to absolute score calibration and small rank gaps.
Kendall rank correlation	Measures pairwise ordering consistency between two rankings.	Gera et al. (2025); Zheng et al. (2023)	Can be unstable when many systems have near-tied performance.
Score correlation	Pearson or Spearman correlation between scalar judge scores and human scores.	Chen et al. (2024a); Wei et al. (2025); Pombal et al. (2025); Chiang et al. (2025)	High correlation does not imply calibrated absolute scores.
Cohen’s Kappa / chance-corrected agreement	Agreement corrected for chance label overlap, especially useful for categorical judgments.	Zheng et al. (2023); Chen et al. (2024b)	Sensitive to label imbalance and annotation protocol.

Table 4 – continued

Metric	Definition / Use	Representative Papers	What It Misses
Consistency rate	Percentage of examples where judgments remain unchanged across perturbations, repeated runs, or original/biased pairs.	Haldar & Hockenmaier (2025); Ye et al. (2025); Yang et al. (2025); Xie et al. (2025)	Does not ensure the consistent answer is correct.
Robustness rate	Fraction of cases where the judge resists a bias-inducing perturbation.	Ye et al. (2025)	Depends heavily on perturbation validity.
Attack success rate	Fraction of cases where a perturbation or adversarial output flips the judge to the wrong verdict or inflates a score.	Chen et al. (2024b); Zheng et al. (2025); Raina et al. (2024)	Attack strength may not reflect natural deployment frequency.
Bias percentage / bias rate	Share of examples where a correct original decision becomes wrong after a bias trigger is added.	Yang et al. (2025); Ye et al. (2025)	Requires a carefully constructed unbiased baseline.
Preference leakage score	Difference in judge preference attributable to relatedness between the synthetic-data generator and evaluator.	Li et al. (2026)	Requires controlled student training and may be hard to estimate for closed models.
Evaluation confidence / uncertainty	Probability mass associated with the judge verdict or score, used to analyze stability and out-of-distribution behavior.	Xie et al. (2025)	Internal token probability need not equal empirical correctness.
Expected calibration error (ECE)	Measures how closely judge confidence matches empirical human agreement.	Jung et al. (2024)	Requires confidence estimates and enough calibration data.
AUROC / AUPRC for failure prediction	Evaluates whether confidence can distinguish likely correct from likely incorrect judgments.	Jung et al. (2024)	Does not itself set a deployment threshold.
Coverage	Fraction of examples evaluated without abstention under selective evaluation.	Jung et al. (2024)	High coverage is undesirable if agreement guarantees fail.
Selective risk / guaranteed agreement	Probability that accepted judge decisions agree with humans above a target threshold.	Jung et al. (2024)	Requires calibration assumptions and may abstain on difficult cases.

Table 4 – continued

Metric	Definition / Use	Representative Papers	What It Misses
Cost per evaluation / tuning cost	Monetary or compute cost of judge calls, hyperparameter search, or large-scale ranking.	Salinas et al. (2025); Wei et al. (2025); Jung et al. (2024)	Low cost alone says nothing about reliability.
Inference latency	Time required to produce judgments, especially in cascaded, checklist, crowd-comparison, or test-time-scaling pipelines.	Yang et al. (2025); Jung et al. (2024); Zhang et al. (2025); Zhou et al. (2025)	Fast judges can still be biased or poorly calibrated.
Best-of-N improvement / extrinsic win rate	Measures whether using a judge to select from candidate generations improves final outputs.	Pombal et al. (2025); Zhang et al. (2025); Zhou et al. (2025)	Improvement may depend on candidate diversity rather than judge quality alone.
Search or refinement gain	Change in task performance when a judge guides beam search or provides a critique for revision.	Zhou et al. (2025)	A positive judge benchmark score may coexist with zero or negative downstream gain.

C.1 Practical Metric Selection

For pairwise open-ended evaluation, the minimum useful report should include human agreement, win/tie rates, position-swapped agreement, and consistency across repeated runs. For system-level leaderboards, rank correlation should be accompanied by uncertainty intervals, adversarial parser checks, and sensitivity to the model pool. For scalar direct assessment, score correlation is not enough: calibration, score variance, ordinal or regression error, and rubric sensitivity should also be reported. Pipelines that use synthetic data should report generator–judge lineage and a preference-leakage audit. When judges guide inference, downstream reranking, search, or refinement gain is the relevant outcome. For high-stakes settings, coverage and guaranteed human agreement make explicit when the judge abstains or escalates.

D Comprehensive Bias Catalogue

Table 5 consolidates the major bias types and reliability failures discussed across the reviewed papers. The purpose is to make the survey usable as a checklist for future LLM-as-a-Judge audits.

Table 5: Biases and reliability failures documented in the reviewed literature.

Bias / Failure	Description	Reported Evidence	Mitigation Direction
Position bias	Preference changes when response order is swapped.	Severe in Wang et al. (2024); included in CALM and RBD.	Balanced position calibration, order averaging, RBD.

Table 5 – continued

Bias / Failure	Description	Reported Evidence	Mitigation Direction
Verbosity bias	Longer answers are favored despite lower correctness or clarity.	Found in MT-Bench-style analysis, CALM, and RBD datasets.	Length-controlled comparisons, checklist grading, bias detection.
Self-enhancement / self-preference	Judge favors outputs generated by itself or same model family.	Included in Ye et al. (2025); discussed as an evaluator fairness risk.	Anonymous model labels, cross-family judges, audit by source.
Preference leakage	A judge favors students trained on synthetic data from the same, inherited, or related model family.	Li et al. (2026) finds systematic preference that grows with relatedness and synthetic-data proportion.	Disclose model lineage, diversify generators and judges, and evaluate with unrelated judge families.
Authority bias	Fake citations, references, or authority cues increase preference.	Chen et al. (2024b) show fake references can attack GPT-4; CALM includes authority perturbations.	Citation verification, source-blind judging, perturbation tests.
Misinformation oversight / fallacy oversight	Judge overlooks factual or reasoning errors when the response appears plausible.	Chen et al. (2024b) and Ye et al. (2025) document failures on factual and reasoning perturbations.	Reference checks, checklist criteria, stronger reasoning prompts.
Gender bias	Judge ignores or tolerates gender-biased content.	Studied in Chen et al. (2024b); LLM judges show vulnerability to gender perturbations.	Safety-aware rubrics, demographic perturbation audits.
Beauty bias	Visually or stylistically appealing content receives preference independent of validity.	Human and LLM judges are vulnerable in Chen et al. (2024b).	Plain-text normalization, style-controlled evaluation.
Bandwagon bias	Judge is swayed by fabricated majority opinion.	Quantified by CALM and RBD; RBD explicitly targets this bias.	Remove social-proof cues, adversarial instruction tests.

Table 5 – continued

Bias / Failure	Description	Reported Evidence	Mitigation Direction
Sentiment bias	Positive or negative tone changes the verdict even when semantics are preserved.	Included in CALM and in Yang et al. (2025).	Tone-controlled perturbations, sentiment-invariant rubrics.
Compassion-fade bias	Evaluation changes with scale or framing of affected groups.	One of CALM’s 12 bias categories.	Explicit fairness criteria and demographic-scale perturbations.
Distraction bias	Irrelevant details pull attention away from core correctness.	Included in CALM’s automated perturbation suite.	Criteria-focused grading and irrelevant-detail filtering.
Diversity / demographic bias	Identity descriptors alter evaluation outcomes.	CALM tests group descriptors such as protected attributes.	Counterfactual demographic evaluation.
Chain-of-thought sensitivity	Verdict changes depending on whether CoT is requested.	Included in CALM; RocketEval finds ordinary CoT alone may not fix lightweight judges.	Structured checklists, evidence templates, CoT audits.
Refinement-aware bias	Telling the judge an answer was refined changes its score.	Included in CALM.	Hide edit history unless it is part of the task.
Self-inconsistency	Same input receives different scores or rankings across repeated runs.	Central result of Haldar & Hockenmaier (2025).	Multiple trials, deterministic decoding, calibration, confidence reporting.
Parsing / template vulnerability	Model output exploits benchmark parser or judge prompt format.	Null models achieve high win rates in Zheng et al. (2025).	Robust parsers, adversarial benchmark tests, hidden evaluation templates.
Universal adversarial trigger	A short suffix appended to many candidate responses inflates absolute scores or flips comparisons.	Raina et al. (2024) demonstrate transfer from surrogate to unseen judges.	Input sanitization, adversarial training, ensemble checks, and comparative formats where appropriate.

Table 5 – continued

Bias / Failure	Description	Reported Evidence	Mitigation Direction
Overconfidence / poor calibration	Judge confidence does not match human agreement probability or shifts across domains.	Jung et al. (2024) find overestimation; Xie et al. (2025) show family-, size-, prompt-, and distribution-dependent confidence.	Uncertainty-aware training, task-specific calibration, selective evaluation, and abstention.